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De tweede verschillen zijn constant, dus de rij is van orde 2.

$\Rightarrow x_n = an^2 + bn + c$

$$\begin{cases} a+b+c=1 \\ 4a+2b+c=2 \\ 9a+3b+c=5 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 4 & 2 & 1 & 2 \\ 9 & 3 & 1 & 5 \end{array}\right]$$

$R_2 \leftarrow R_2 - 4R_1$, $R_3 \leftarrow R_3 - 9R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -3 & -2 \\ 0 & -6 & -8 & -4 \end{array}\right]$$

$R_3 \leftarrow R_3 - 3R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -3 & -2 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$\Rightarrow \boxed{c = 2}$

$-2b - 3(2) = -2$

$\Rightarrow \boxed{b = -2}$

$a + b + c = 1$

$\Rightarrow \boxed{a = 1}$

$\Rightarrow \boxed{x_n = n^2 - 2n + 2}$

$x_8 = 8^2 - 2 \cdot 8 + 2 = 50$

References